

and XOR. These logical operators are used together in complex ways to determine if a particular condition is True or False. Examples:

`x AND y` returns the value True if the condition is true for both `x` and `y` or else it returns False

`NOT x` returns the value True if `x` is false (or null) and False if `x` is true

`x OR y` returns the value True if either or both `x` and `y` are true. It only returns false if both `x` and `y` are False for the given condition

`x XOR y` returns the value True only if either `x` or `y` are true. If both `x` and `y` are true then it will return the value False.

Using Boolean operators is not restricted to computer programming. End-users also use Boolean expressions when they need to filter results in a search engine window. For example, the Boolean expression “computers AND hardware NOT memory” when used in a search engine will return all pages containing the words “computers” and “hardware” while excluding all pages containing the word “memory”.

6.3.10 C/C++

As programming languages go, C and C++ (pronounced C plus plus) are among the older languages that are still going strong. C was originally developed in the mid 1970s, and was primarily used to write programs for the UNIX operating system. Since then, its use has expanded, and nowadays it is used to write applications for nearly every available system. Those working on embedded technologies and hardware interfaces are particularly enamoured by C due to its flexibility and efficient use of memory. C++ is an offshoot of the C language with almost identical syntax. The main improvement that C++ has over C is that it supports object-oriented features that allow programmers great leaps in productivity and efficiency.

6.3.11 Character

Any number, alphabetical character, symbol, or punctuation mark is a character. A computer assigns one byte for each character.